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Test Cases
input 1 : A = [1, 3, 5]
Output 1 : 8
input 2 : A = [2, 3]
Output 2 : 2
We define of [X, Y] as number of different corresponding bits in the binary representation of X & Y. for example, f(2,7) = 2, since the binary representation of 2 & 7 are 010 & 111, respectively. The first & the third bit differ, so f(2,7) = 2.

You are given an array of N positive integers, $A_1, A_2, \dots A_n$. Find sum of $f(A_i, A_j)$ for all pairs (i, j) such that $1 \le i, j \le N$. Return the answer modulo $10^9 + 7$.

⇒ Given
f(x,y) = no. of different bits in binary of x & y (checking every bit.
→ TC = (N^2) → TLE
So, to solve this we can use the concept of bits manipulation → by checking the bits of x & y.

Lets look at a particular kth bit.
if kth bit x is 1 & kth bit of y is 0.
or vice versa.
so ans = 2 for pair (x,y) & (y,x).
↳ let cnt = 1 nos of having bit k=1
cnt = 2 no.s of having bit k=0.
Contribution 2 × count 1 × count 0.
for (x,y) & (y,x)

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Step to Solve

For every bit (0 → 31)

i) Count members having that bit set.

ii) Compute pairs.

iii) Add to answer

Code

```
long MOD = 100000007;
long totalPairs(ArrayList<Integer> arr) {
    long n = arr.size();
    long ans = 0;
    for (int bit = 0; bit < 32; bit++) {
        long cnt1 = 0
        for (int i = 0; i < n; i++) {
            if (arr.get(i) & (1 << bit))
                Count++;
        }
    }
    long contribution = (cnt1 * cnt0)
    contribution = (2 * contribution)
    ans = (ans + contribution)
} return ans;
}
```